

Foundations of Electricity

A short course for beginners

Chapter 1: What Is Electricity?

Electricity is the flow of electric charge, usually carried by electrons moving through a conductor such as a copper wire. Materials that let charge move freely are called conductors; materials that resist this movement strongly are called insulators.

An atom's electrons can be dislodged from their outer shell and pushed from atom to atom. This directed movement of charge, when it happens in a continuous loop called a circuit, is what we call an electric current.

Static electricity, by contrast, is charge that has built up on a surface and is not flowing anywhere — the small shock from touching a doorknob after walking on carpet is static charge suddenly discharging, not a circuit's current.

Chapter 2: Voltage and Circuits

Voltage, also called potential difference, is the energy given to each unit of charge by a source such as a battery, measured in volts. It is voltage that pushes current around a circuit, in the same way that pressure pushes water through a pipe.

A circuit needs a closed, unbroken path: a source, a conductor, and a load (something that uses the electrical energy, such as a bulb or a motor). Break the path anywhere — for instance by opening a switch — and the current stops everywhere in that loop at once.

In a series circuit, components are connected one after another so the same current passes through each of them, while the source voltage is shared out across them. In a parallel circuit, components share the same two connection points, so each sees the full source voltage, while the total current from the source splits between the branches.

Chapter 3: Electrical Power, Energy, and Household Safety

Electric power is the rate at which electrical energy is converted — into heat, light, or motion — measured in watts. Power is the product of voltage and current: $P = V * I$. Combining this with Ohm's Law ($V = I * R$) also gives $P = I^2 * R$ and $P = V^2 / R$, three equivalent ways to reach the same answer depending on which two quantities you already know.

Worked example: a household kettle is rated at 230 volts and draws 4 amperes while heating. Its power is $P = V * I = 230 * 4 = 920$ watts, meaning it converts about 920 joules of electrical energy into heat every second.

Electrical energy is power sustained over time: energy equals power multiplied by time. Utility bills measure this in kilowatt-hours (kWh), not joules — one kWh is the energy used by a 1000-watt appliance running for one hour. That 920-watt kettle run for 15 minutes (0.25 hours) uses $0.920 * 0.25 = 0.23$ kWh.

Because power rises with the SQUARE of current ($P = I^2 * R$), a fault that lets current run higher than a circuit's wiring was designed for generates heat far faster than the current increase alone suggests — this is precisely why household circuits are protected.

A fuse is a deliberately weak link — a thin wire or metal strip inside it melts and breaks the circuit once current exceeds a rated value, before the wiring elsewhere in the house can overheat. A circuit breaker does the same job with a mechanical switch that trips instead of melting, so it can be reset and reused rather than replaced.

Earthing (grounding) connects a device's metal casing to the ground via a low-resistance wire. If a fault lets a live wire touch the casing, earthing gives the resulting fault current a safe, low-resistance path to flow through instead of through a person who touches the casing, and that surge of current is usually large enough to blow the fuse or trip the breaker immediately.

Chapter 4: Resistance and Ohm's Law

Resistance is a material's opposition to the flow of current, measured in ohms. Thin, long, or poorly conducting wires have more resistance than thick, short, highly conductive ones — this is why a hairdryer's heating element is a thin coiled wire, not a thick one.

Ohm's Law relates the three quantities: voltage equals current multiplied by resistance, written $V = I * R$. Equivalently, current equals voltage divided by resistance: $I = V / R$.

A common misconception is that increasing resistance increases current, because both words sound like they mean 'more'. Ohm's Law says the opposite: if voltage is held constant and resistance goes up, current goes DOWN, because $I = V / R$ and a larger denominator gives a smaller result.

Worked example: a 12 volt battery is connected across a 4 ohm resistor. The current is $12 / 4 = 3$ amperes. If that resistor is swapped for a 6 ohm resistor at the same 12 volts, the current falls to $12 / 6 = 2$ amperes — less current, not more, because resistance rose.

Chapter 5: Magnetism and Electromagnetic Induction

A current-carrying wire produces a magnetic field around itself — this is the basis of electromagnets, which are simply coils of wire that become magnetic only while current flows through them, unlike a permanent magnet.

Electromagnetic induction, discovered by Michael Faraday, is the production of a voltage in a conductor when the magnetic field around it changes. This is the principle behind electric generators, which convert mechanical motion into electrical energy.